Ex: NO: 1 | Study of various Network Commands used in Linux and Windows

Aim:

To study of various network commands used in Linux and windows.

Commands:

1) arp -a:

ARP is short form of address resolution protocol it will show the IP address of your computer along with IP address and MAC address of your router

Output:

Internet Address	Physical address
172.16.8.149	172:16:8:13:f3:fe
172.16.9.212	7c:57:58:38:da:a3
172.16.8.186	7c:57:58:38:e6:24
172.16.8.249	00:27:0e:13:f5:77
172.16.11.207	7c:57:58:38:46:87

2) hostname:

This is the simplest of all TCP/IP commands. It simplify displays the name of your computer.

Output:

localhost . local domain

3) ipconfig /all:

This command displays detailed configuration information about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of ethernet adopter in your system.

Output:

DHCP server : 192.168.112.254
DHCPV6 IAID : 58722 3126
DHCPV6 Client DUID : 00-01-00-01-2D-D7-41
-B3-7C-57-58
primary WINS Server : 192.168.112.2
NetBIOS Over Tcpip : Enabled

4) Netstat : (Network statistics)

netstat displays a variety of statistics about a computers active TCP/IP connections. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing routing tables, interface statistics etc.

netstat -r

Output:

Interface List
2. 7C 57 58 35 10 fb
17. 00 50 56 C0 00 01
11. 00 50 56 C0 00 08
1 Software LoopBack Interface 1.

5) nslookup: (name server lookup) is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer, the MX records for a domain or the NS servers of a domain nslookup can operates in two modes. interactive and non-interactive.

Eg: nslookup www.google.com

Output:

Server unknown

Address : 172.16.8.1
Non-authoritative answer
Name : Star mini or google.com
Address : 2903 2880 f137 182; 6a3e : b00c :
0 : 25de 157.240.192.35
Aliases www.goggle.com

Linux commands :

1) arp -a :
gateway (172.16.8.1) at 7c 5a 1c cf be 45
[ether] on enp250

2) ifconfig :
enp250 : flags = 4163 <UP, BROADCAST,
RUNNING, MULTICAST> mtu 1500
inet 172.16.8.112 netmask 255 255 252
broadcast 172.16.11.255
ether 50 : 9a : 4c : 35 : 0f : 45
txqueuelen 1000 (Ethernet)

Wlp350 : flags = 4099 < UP, BROADCAST, MULTICAST >
mtu 1500
ether 4a : 25 : d4 : c1 : a3 : ab
txqueuelen 1000 (Ethernet)

3) hostname
localhost localdomain

4) nmblookup -A 172.16.8.1
Looking up status of 172.16.8.1
NO reply from 172.16.8.1

5) netstat -r
Kernal IP routing table
Destination Gateway Genmask
Flags MSS in Window
default gateway 0000 UG 0000 172.16.8.0
0.0.0.0 255.255.255.0 U 00 00

Important Linux Networking Command

1) ip:
Every administrator will need in daily work from setting up new system
Eg: ip address show
link (lookup 00:00:00:00:00:00 bnd
00:00:00:00:00:00)
inet 127.0.0.1 18 Scope host 10.

2) intr:
Serves as a network diagnostic tool.
mtr google.com
Packets
loss%. Snt Last avg Boot ists
0.0%. 17.8 0.1 0.1 0.0 0%.

3) tcpdump: designed for capturing and displaying packets
Output: tcpdump - p

4) Ping: Tools to verify IP level connectivity to another TCP/IP computer by sending TCP/IP Internet control message protocol Echo request messages.
Output: Ping local host
ping localhost c local host (::1) 56
data bytes 64 bytes from localhost
(::1) time = 0.053 ms

Observation:

1. Which command is used to find the reachability of a ~~to~~ host machine from your device?

Ping command <hostname /IP> ping is primary TCP command used to troubleshoot connectivity reachability.

2. Which command will be give the details of hops taken by a packet to reach its destination?

mtr (Matt's trace route)

It show route from a computer to a specified host mtr provides a lot of statistics about each hop such as response time and percentage.

3. Which command displays the ip configuration of your machine?

ip <options> <objects> <command>

ip command can show address information manipulate routing plus display network various devices network various devices interfaces & funnels.

4. Which command displays the TCP port status in your machine?

net stat

It displays variety of statistic about a computer active TCP/IP connection

5. Write the modify the ip configuration in a linux machine?

* Assigning IP address to interfaces ip address add 192.168.1.254124
* Deleting IP address
  ip address del 192.168.1.254124
  dev enp803

Result:

Thus, the network commands using Linux and Windows are successfully executed and the output is verified.